



Project Prism: Duplicate Project Checker

Project Prism is a web-based platform designed to prevent the repetition of academic projects by students within a university or institution.



by Cheppali Kalyani

Project Prism: Key Features

1 Project Data Collection

Students submit project details, including title, abstract, guide, and panel information.

2 Duplicate Detection

The system automatically checks for project similarity using natural language processing (NLP).

3 Project Evaluation Panel

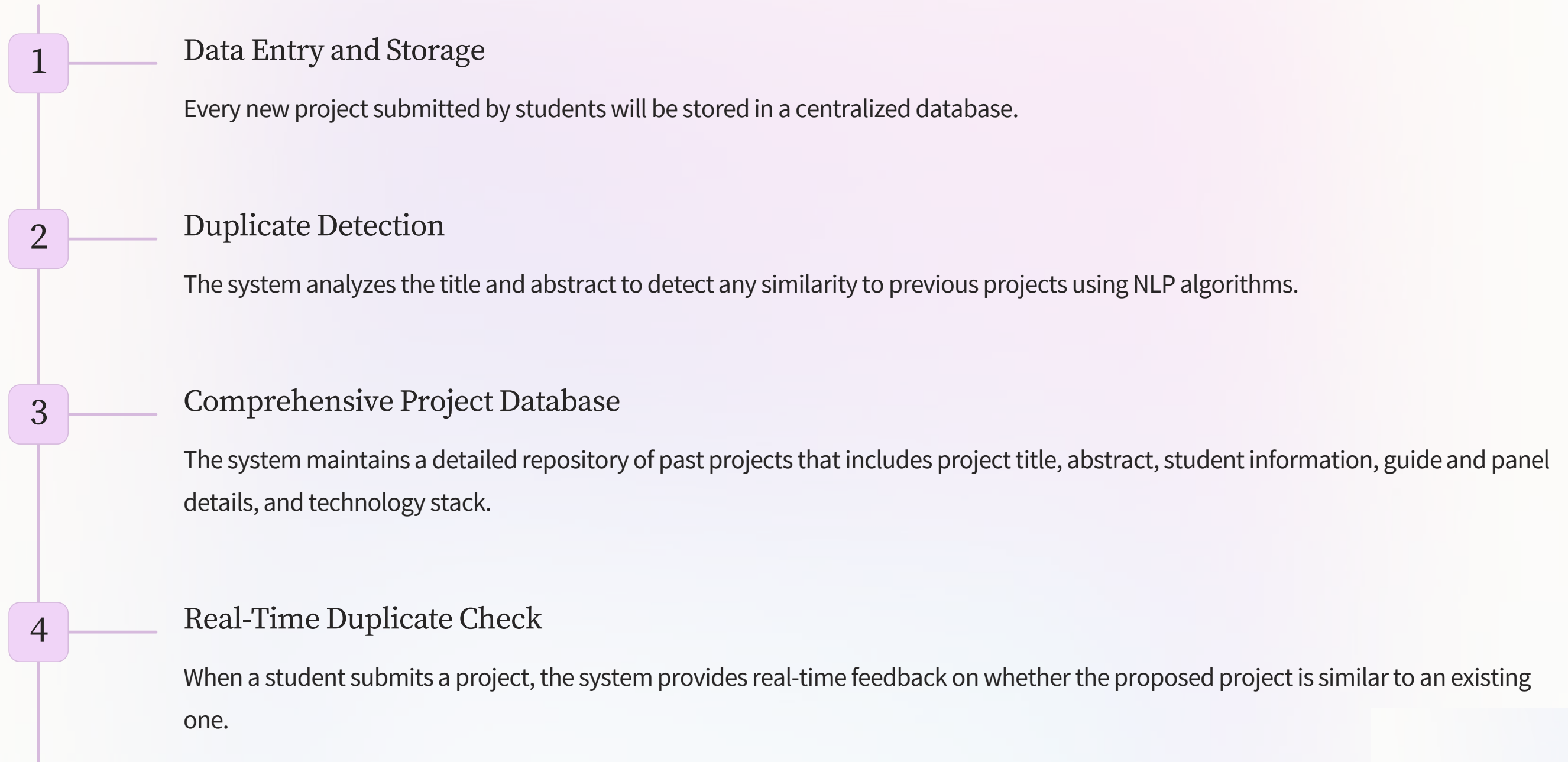
The platform keeps track of the panel that evaluates the project, ensuring transparency.

4 Project Search and Reporting

Students and faculty can search the database to review past projects by department, technology, or year.

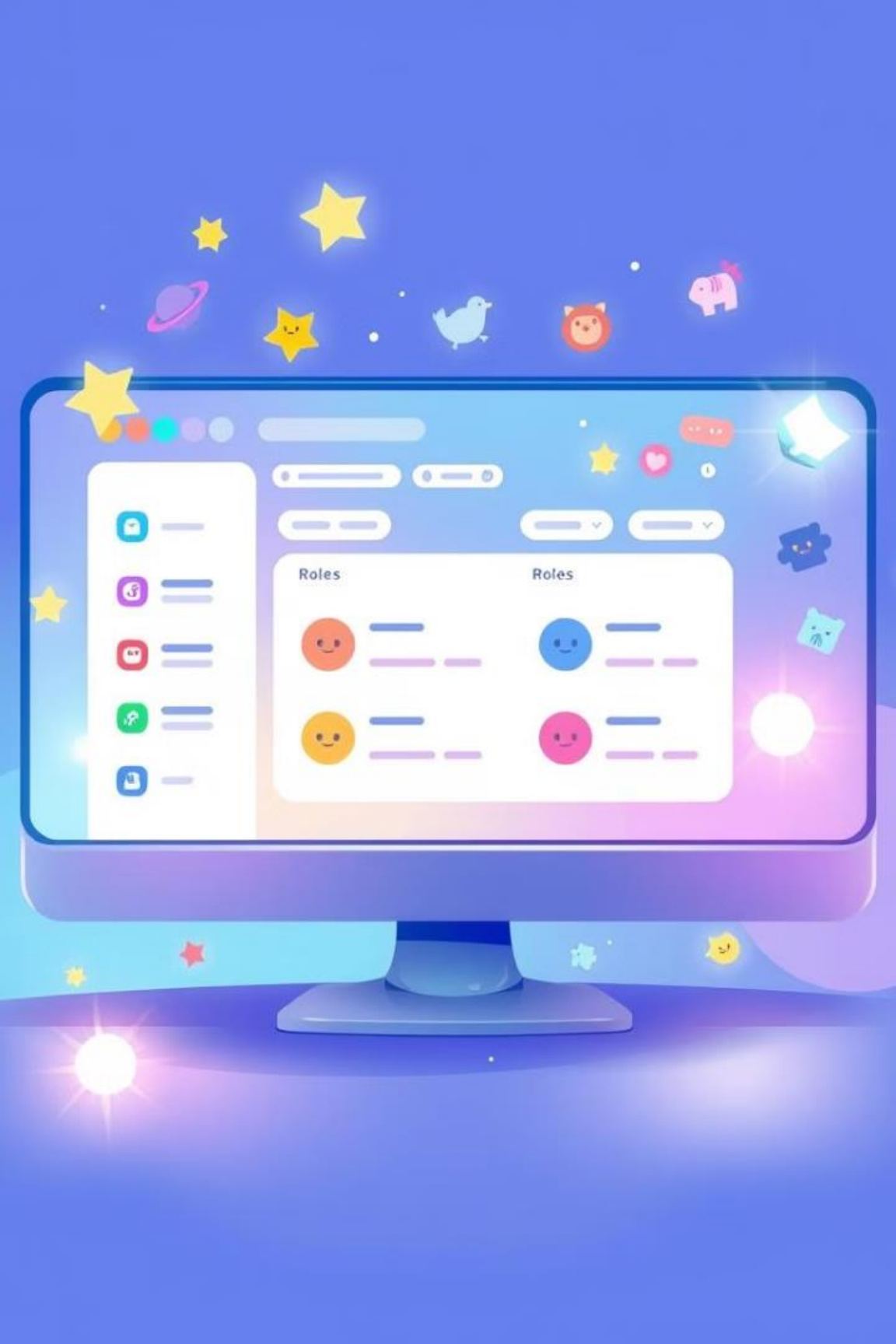


Problem Solving Approach



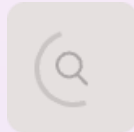
User Roles and Permissions

Role	Permissions
Students	Submit projects, check for duplicates
Guides	Review proposed projects, ensure alignment with university standards
Administrators	Manage the project database, panel assignments, issue notifications





Alerts and Notifications



Duplicate Detection Alerts

If a duplicate project is detected, the student, guide, and panel members will be notified through email or SMS.



Project Approval Notifications

Status updates regarding project approvals or rejections will also be shared with relevant parties.

Technologies to be Used

Frontend

- HTML, CSS, JavaScript
- Bootstrap

Backend

- Python Flask
- NLP Libraries (spaCy, NLTK)
- TF-IDF, Cosine Similarity,
Levenshtein Distance

Database

- MySQL/PostgreSQL

Functionalities in the System



How I Plan to Solve the Problem

User-Friendly Interface

The project form will be easy to use, ensuring that students provide complete and accurate information.

Efficient Algorithms

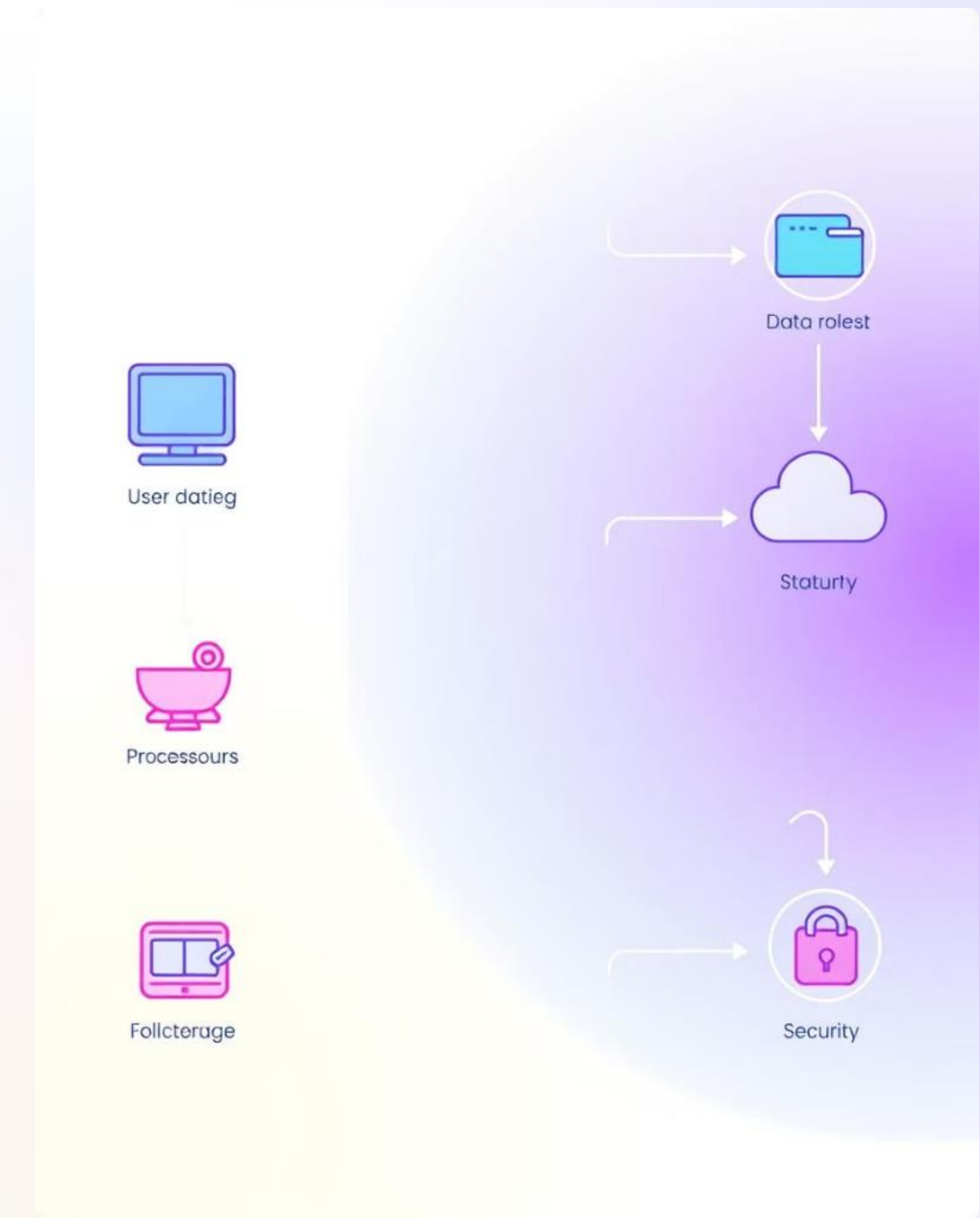
I will integrate efficient NLP algorithms like TF-IDF, Cosine Similarity, and Levenshtein Distance to ensure that the duplicate detection system accurately identifies projects with significant overlap.

Data-Driven Decisions

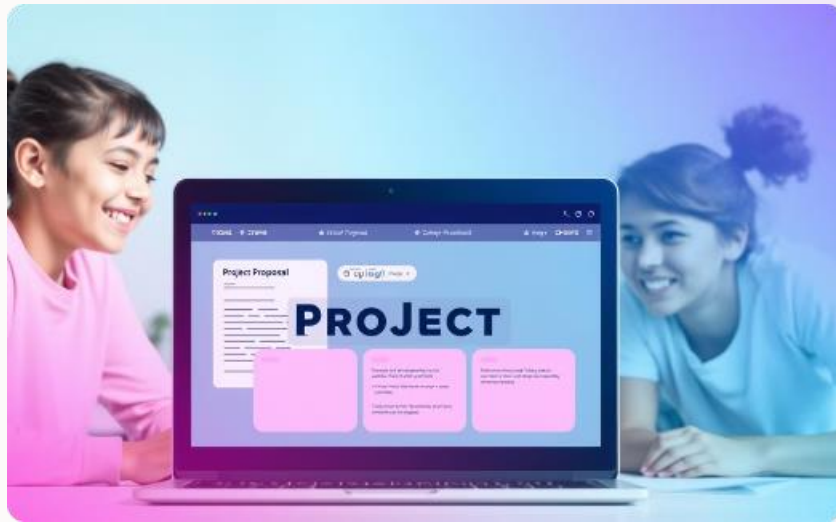
Guides and evaluators will have access to a complete project repository, helping them make better decisions and giving students ideas for innovation.

Scalability

By using cloud technologies like AWS S3 and PostgreSQL, the system will be scalable and capable of handling large volumes of data across multiple departments.



Benefits of Project Prism



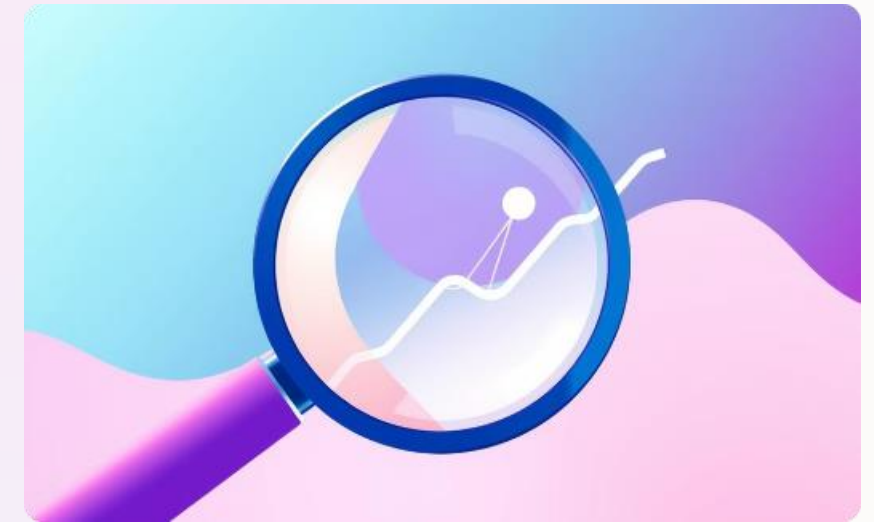
Reduced Project Duplication

Students will be forced to come up with new, innovative ideas, improving the quality of projects.



Streamlined Process

The project submission and approval process will be faster and more transparent.



Better Resource Management

The university can better track which projects are being done and identify trends in research.



Conclusion

Project Prism is an innovative solution to the problem of project duplication in academic institutions. By combining natural language processing, web technologies, and a user-friendly interface, this project will help universities maintain a repository of unique and original projects, promote creativity, and ensure fairness in project evaluation.